

Data Abstraction & 3-Level Schema Architecture

Q. What is Data Abstraction in a DBMS? Explain the three levels of data abstraction with a suitable example.

> **Definition to Remember**

1. The Three Levels of Data Abstraction

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- VIEW LEVEL (External) End Users see their specific views
- LOGICAL LEVEL (Conceptual) Designers see the full structure
- PHYSICAL LEVEL (Internal) DBA sees raw storage details

...

1. Physical Level (Internal Level) *Lowest Level*

2. Logical Level (Conceptual Level) *Middle Level*

3. View Level (External Level) *Highest Level*

2. Example University Database

| Level | What is Seen |

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

> `Phy